# EKF-Based 2D SLAM

- Differential-drive robot with range-bearing landmarks

- Jointly estimate robot pose and an initially unknown map

- Student: _____

# Problem and scope

- Odometry accumulates drift; the map is initially unknown.
- 10 stationary landmarks; known IDs; 360 deg sensing to 4.5 m.
- Known starting pose; Gaussian simulated noise.
- Python implementation plus Gazebo Classic / ROS 2 demo.

# Joint state and uncertainty

- X = [x, y, heading, m1x, m1y, ..., mNx, mNy]

- 23 states after all 10 landmarks are discovered.

- Full covariance preserves robot-landmark correlations.

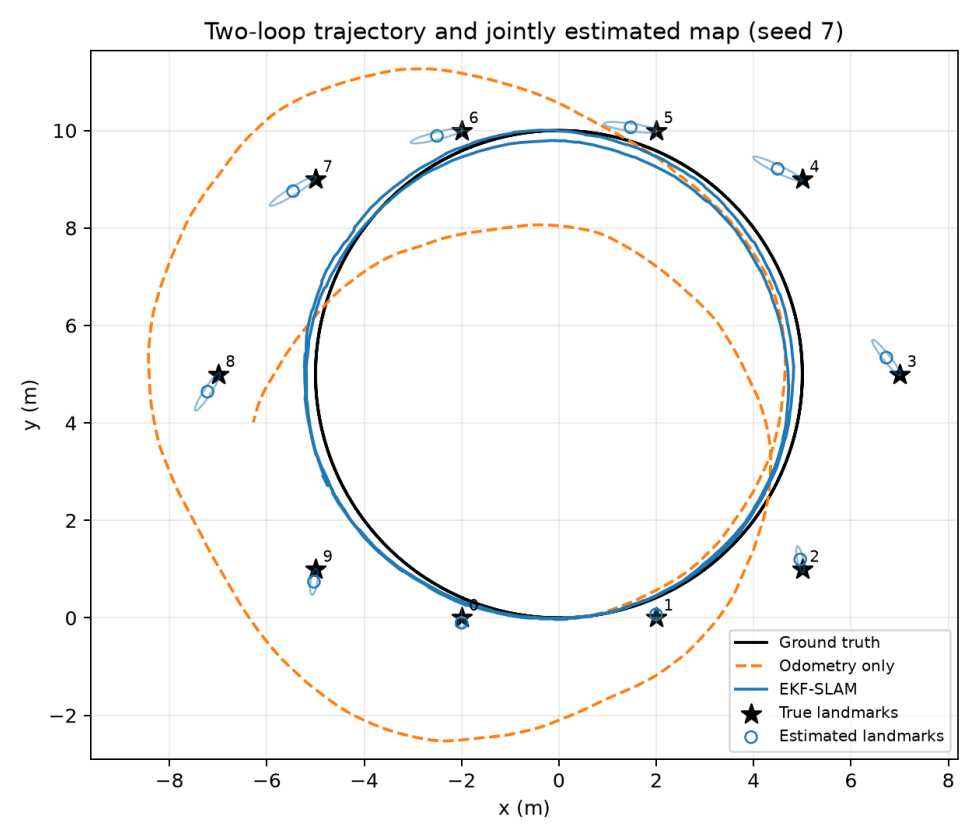
- The filter receives no true landmark coordinates.

# Prediction and correction

- Predict pose from left/right wheel travel; propagate covariance.
- First sighting: inverse sensor model + correlated augmentation.
- Repeat sighting: range/bearing innovation and Kalman gain.
- Wrap angles; linear solves; Joseph covariance update.

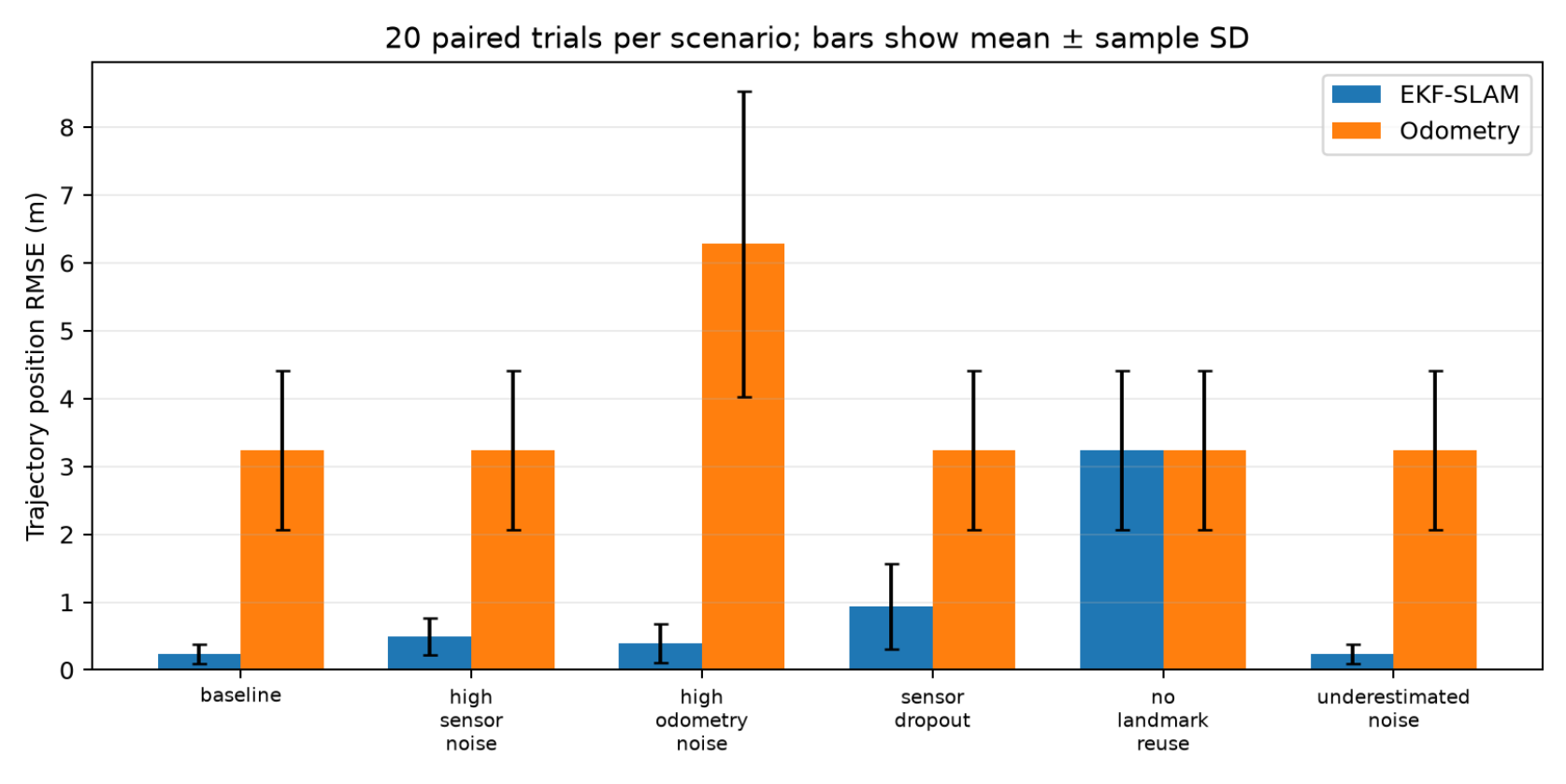
# Trajectory and map

- Black: truth; blue: EKF-SLAM; dashed orange: odometry.
- Illustration uses seed 7, not the trial average.



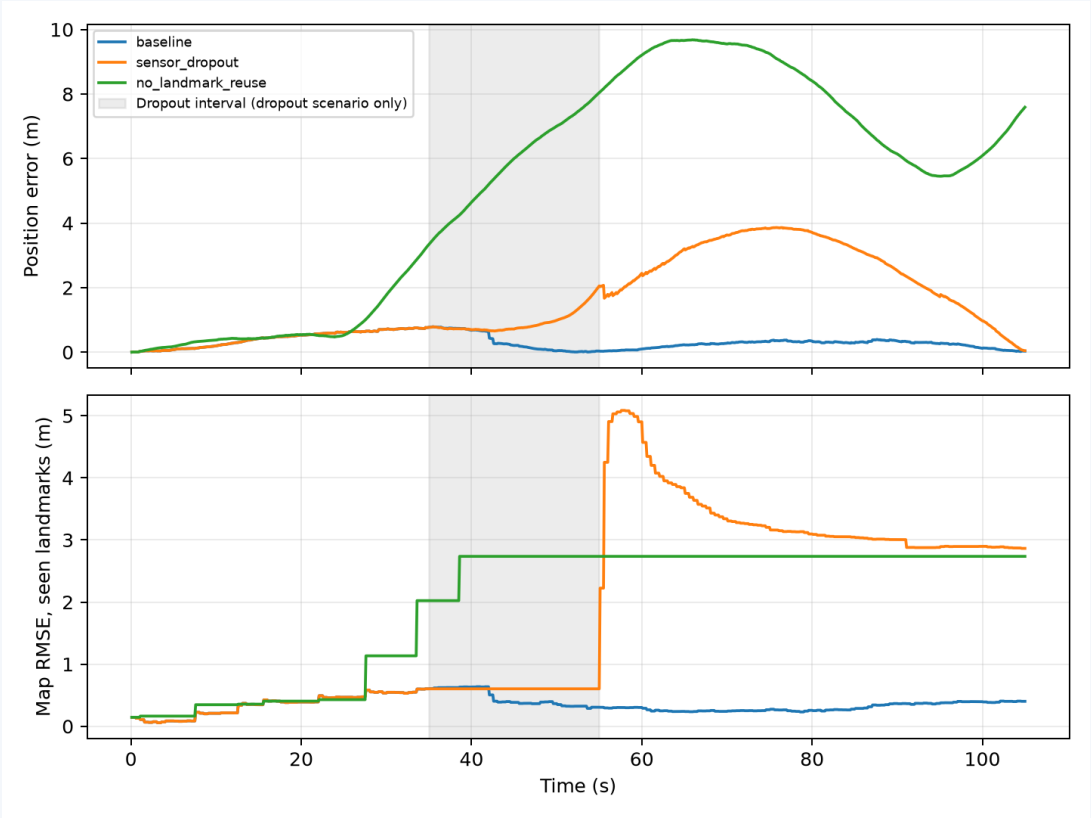
# Paired Monte Carlo results

- 20 runs per scenario: EKF RMSE 0.238 m; odometry 3.233 m.
- Error bars show sample standard deviation across runs.



# What happens during an outage?

- No landmark corrections between 35 and 55 s.
- Robot position can recover while the map stays distorted.
- No-reuse ablation removes every repeated observation.



# Accuracy is not confidence

- Scale all assumed noise standard deviations by 0.35.

- Trajectory RMSE stays about 0.238 m.

- Mean NIS rises from 2.02 to 16.50: overconfident uncertainty.

- Mean NIS alone does not establish full SLAM consistency.

# Gazebo demonstration

- Gazebo Classic 11 + ROS 2 Humble in devbox.

- Encoder-mode odometry with added wheel noise.

- Synthetic range/bearing from world poses; known IDs.

- Completed headless physics run; GUI launcher supplied.

- Physics-run metrics are separate from Monte Carlo results.

# Conclusions and limitations

- Landmark reuse substantially reduces odometry drift.

- Outages and nonlinear errors can distort the map.

- Eleven automated tests and reproducible experiment outputs.

- Limitations: known IDs, ideal visibility, small map.

- Next: sensor extraction, robust association, more trajectories.